Scalable Geometric Semi-Supervised Learning: Empirical Baseline of Manifold Regularization

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Motivation

When labels are expensive, geometry is free

Most classification problems have abundant *unlabeled* data and very few *labeled* examples. Photographs, sensor readings, scientific events — we collect millions of examples cheaply but can afford to label only a few hundred. **Semi-supervised learning (SSL)** closes this gap by using the *shape* of the unlabeled data to guide the classifier.

The classical theoretical framework — **manifold regularization** (Belkin, Niyogi & Sindhwani, 2006) — has been mathematically proven for two decades. But it requires inverting a dense matrix the size of *every data point*, costing O(N^3) time and O(N^2) memory. That wall is what has kept the technique out of large-scale practice.

This poster is the **first phase** of a multi-phase study testing whether a research-stage engine, **Lumawarp** (Lucidity Sciences), can preserve the *theory* while removing the *scaling wall*. Phase 1 reproduces and characterizes the classical baseline; Phase 2 (in preparation) runs the Lumawarp comparison.

Theoretical Framework

The Laplacian as geometric bridge. *Belkin & Niyogi, Laplacian Eigenmaps* (2003) showed that the graph Laplacian $L = D - W$ — built from a k-NN graph with heat-kernel edge weights — discretely approximates the Laplace–Beltrami operator Δ_M on the underlying data manifold. Its eigenfunctions define the manifold's natural coordinate system; level sets of the smoothest such functions are the geometrically *correct* classification boundaries. Minimizing the Dirichlet energy $f^T L f$ finds functions that vary slowly along the manifold.

From eigenmaps to a classification framework. *Belkin, Niyogi & Sindhwani, Manifold Regularization* (2006) added $f^T L f$ as a regularization penalty alongside the standard RKHS norm:

$$f^* = \underset{f \in \mathcal{H}_K}{\operatorname{argmin}} \frac{1}{l} \sum_{i=1}^{l} V(x_i, x_i, f) + \gamma_A ||f||_K^2 + \gamma_I f^T L f$$

The Representer Theorem expands f^* over all $l + u$ data points, yielding the O(N^3) time and O(N^2) memory costs that bound classical practice — and the scaling wall this study addresses. Belkin himself flags this as the natural future-work direction: *"a small set of basis functions"*.

Kernel warping — absorbing geometry into the inner product. *Sindhwani, Niyogi & Belkin, Beyond the Point Cloud* (2005) provides the alternative blueprint. Rather than adding $f^T L f$ as an explicit penalty, *deform the kernel itself* using the Laplacian:

$$\tilde{k}(x,z) = k(x,z) - k_x^T (I + MK)^{-1} M k_z, M = L^p$$

The warped kernel bakes manifold geometry into the RKHS inner product directly. Standard supervised methods (SVM, RLS) can then be applied in the warped RKHS using only labeled data — the unlabeled geometry has been absorbed. This is the architectural family Lumawarp belongs to.

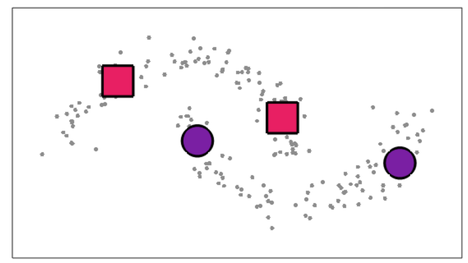
Implementation

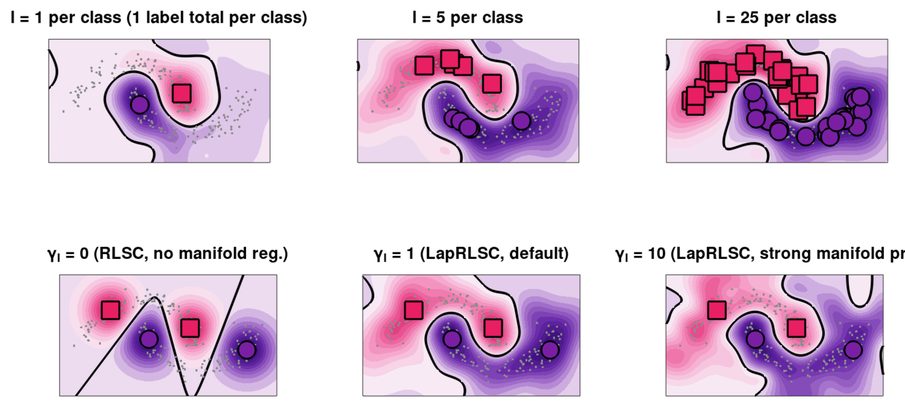
Method A — Classical baseline: ManifoldLearn (Belkin et al., 2006)
The reference MATLAB toolbox from the original paper. Builds a graph Laplacian over all N points, then solves a dense $N \times N$ linear system. Mathematically clean, computationally heavy.

Method B — Lumawarp (Lucidity Sciences)
A research-stage kernel-based ML engine that operates in the same theoretical family — kernel methods that exploit unlabeled-data geometry — but addresses the scaling wall through an architectural innovation: **it condenses the full graph Laplacian into a compact set of *manifold support vectors***, exposing only one user-facing knob — **NSV (k)**, the number of support vectors retained — which controls how compact the manifold representation is.
Storage drops from O(N^2) to O(kN).

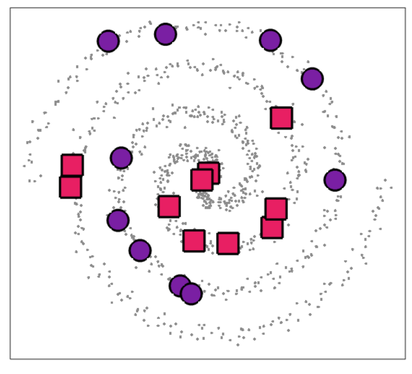
Experiments

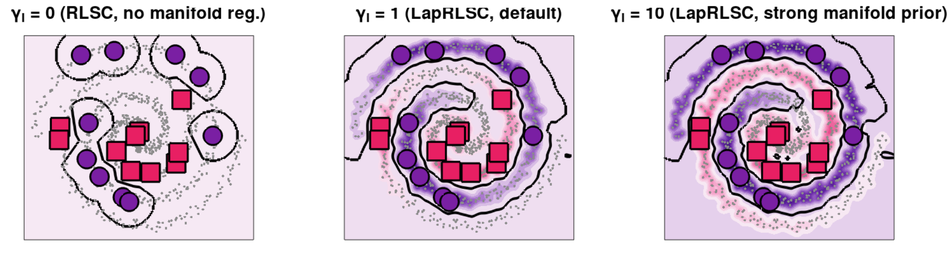
Two-moons dataset
196 unlabeled (gray) + 2 labeled per class



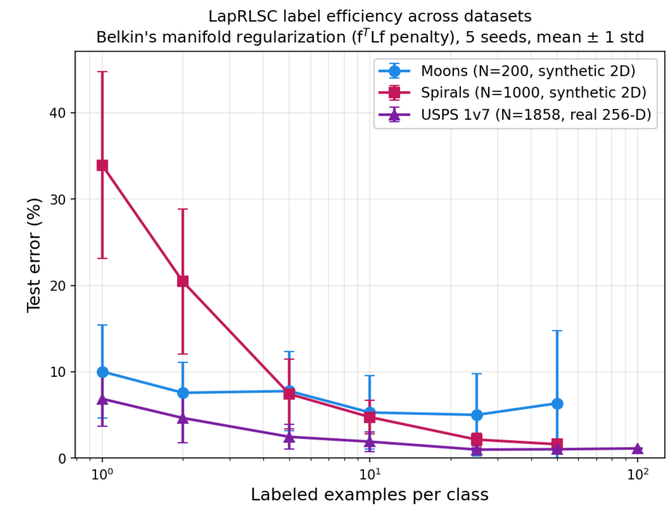


Two-spirals dataset
980 unlabeled (gray) + 10 labeled per class





Results



The cross-dataset findings are as follows:
(1) The manifold regularizer is doing real work. The γ_I sweep on moons (Panel 6) drops test error from 12% → 0% with the same 4 labels by turning on $f^T L f$. The same sweep on spirals (Panel 9) drops error from 42% → 17% — a 25-point gain. The boxed term in Belkin's objective is not theoretical decoration; *it carries the classifier* on data with non-trivial manifold structure.
(2) The classifier needs very few labels once γ_I is on. With $\gamma_I = 1$ fixed, the moons label sweep (Panel 7) recovers a near-perfect decision boundary using **one** labeled example per class (1% error). The classifier is essentially inferring class identity from the geometry of the unlabeled cloud alone.
(3) The story generalizes to real high-dimensional data. The cross-dataset curve above shows the same machinery — same Octave toolbox, same kernel-SSL framework — handles 2D moons, harder 2D spirals, *and* real 256-dimensional USPS handwritten digits. **Headline number: USPS at 1 label per class → 6.9% test error.** Manifold-aware SSL extracts almost all the signal from real image data with a single labeled example per class.

Ongoing Study/Future Work

From classical reproduction, to scalable comparison

Belkin's manifold-regularization term penalizes f for varying along the data manifold encoded by L, the graph Laplacian over all $l + u$ points. **The classical wall.** The Representer Theorem expands f^* over all $l + u$ data points → an $(l + u) * (l + u)$ linear-system solve → O(N^3) time and O(N^2) memory.
Lumawarp's solution. Condense L into a compact set of *manifold support vectors*, controlled by the NSV (k) knob. Storage drops to O(kN).
Phase 2 — Experimental Program:
k-sweep on shared splits. NSV ∈ {8, 16, …, 512} across moons, spirals, USPS at $N \in 200, \ldots, 20000$ on the same Python- generated splits used in this poster.
Saturation curve $k^*(N)$. Smallest k at which Lumawarp accuracy plateaus. **Hypothesis: $k^*(N) \sim \sqrt{N}$** which would empirically explain an $O(N^3) \rightarrow O(N^2)$ time-complexity bridge.
Accuracy-equivalence test. Confirm Lumawarp at k^* matches the LapRLSC error rates on every dataset where both can run.
HEP scalability demonstration. Run Lumawarp on HIGGS and SUSY ($N \approx 10^6$)— scales where classical Belkin cannot run. Comparison framing: "biggest tractable Belkin vs full-scale Lumawarp."
Quality–cost surface. Map test error as a function of (labels/class) × (NSV) × (N).